



Oracle Database Administration

I

Chapter 7: Users and Basic Security

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Objectives:

1. Create and manage database user accounts:
 - Authenticate users
 - Assign default storage areas (tablespaces)
2. Grant and revoke privileges
3. Create and manage roles
4. Create and manage profiles:
 - Implement standard password security features
 - Control resource usage by users

Introduction:

The following terms relate to administering database users and assist you in understanding the objectives:

- A *database user account* is a way to organize the ownership of and access to database objects
- A *password* is an authentication by the Oracle database
- A *privilege* is a right to execute a particular type of SQL statement or to access another user's object
- A *role* is a named group of related privileges that are granted to users or to other roles
- *Profiles* impose a named set of resource limits on database usage and instance resources, and manage account status and password management rules.
- A *quota* is a space allowance in a given tablespace. This is one of the ways by which you can control resource usage by users.

1. Database User Accounts:

Each database user account has:

- A unique username
- An authentication method
- A default tablespace
- A temporary tablespace
- A user profile
- An initial consumer group
- An account status

A schema:

- Is a collection of database objects that are owned by a database user
- Has the same name as the user account

Each database user account has:

- **A unique username:** Usernames cannot exceed 30 bytes, cannot contain special characters, and must start with a letter.
- **An authentication method:** The most common authentication method is a password. Oracle Database supports password, global, and external authentication methods (such as biometric, certificate, and token authentication).
- **A default tablespace:** This is a place where a user creates objects if the user does not specify some other tablespace. Note that having a default tablespace does not imply that the user has the privilege of creating objects in that tablespace, nor does the user have a quota of space in that tablespace in which to create objects. Both of these are granted separately.
- **A temporary tablespace:** This is a place where temporary objects, such as sorts and temporary tables, are created on behalf of the user by the instance. No quota is applied to temporary tablespaces.
- **A user profile:** This is a set of resource and password restrictions assigned to the user.
- **An initial consumer group:** This is used by the Resource Manager.

- **An account status:** Users can access only “open” accounts. The `account_status` may be:
 - OPEN: The account is available for use.
 - LOCKED: This indicates that the DBA deliberately locked the account. No user can connect to a locked account.
 - EXPIRED: This indicates that the lifetime has expired. Passwords can have a limited lifetime. No user can connect to an EXPIRED account until the password is reset.
 - EXPIRED & LOCKED: Not only has the account been locked, but its password has also expired.
 - EXPIRED (GRACE): This indicates that the grace period is in effect. A password need not expire immediately when its lifetime ends; it may be configured with a grace period during which users connecting to the account have the opportunity to change the password.
 - LOCKED (TIMED): This indicates that the account is locked because of failed login attempts. An account can be configured to lock automatically for a period after an incorrect password is presented a certain number of times.
 - EXPIRED & LOCKED (TIMED)
 - EXPIRED (GRACE) & LOCKED
 - EXPIRED (GRACE) & LOCKED (TIMED)

A schema:

- Is a collection of database objects that are owned by a database user
- Has the same name as the user account

Schema objects are the logical structures that directly refer to the database’s data. Schema objects include such structures as tables, views, sequences, stored procedures, synonyms, indexes, clusters, and database links. In general, schema objects include everything that your application creates in the database.

2. Managing Default Users:

When you create a database, Oracle creates several default database users. These default users would either set a default password or take input from database creation. With Oracle 19c, all of the default accounts are locked on installation except for `sys` and `SYSTEM`. You can view the default user accounts, as follows:

```
SQL> select username from dba_users order by 1;
```

Note: If you are working in a pluggable database environment, you can view all users while connected as a privileged account to the root container by querying `CDB_USERS`.

2.1. Predefined Administrative Accounts:

- `SYS`:
 - Owns the data dictionary and the Automatic Workload Repository (AWR)
 - Used for startup and shutdown of the database instance
- `SYSTEM`: Owns additional administrative tables and views
- `SYSBACKUP`: Facilitates Oracle Recovery Manager (RMAN) backup and recovery operations
- `SYSDG`: Facilitates Oracle Data Guard operations
- `SYSKM`: Facilitates Transparent Data Encryption wallet

Operations

The `SYS` and `SYSTEM` accounts have the database administrator (DBA) role granted to them by default. In addition, the `SYS` account has all privileges with `ADMIN OPTION` and owns the data dictionary. To connect to the `SYS` account, you must use the `AS SYSDBA` clause for a database instance and `AS SYSASM` for an Automatic Storage Management (ASM) instance.

Any user that is granted the `SYSDBA` privilege can connect to the `SYS` account by using the `AS SYSDBA` clause. Only “privileged” users who are granted the `SYSDBA`, `SYSOPER`, or `SYSASM` privileges are allowed to start up and shut down instances. The `SYSTEM` account does not have the `SYSDBA` privilege. `SYSTEM` is also granted the

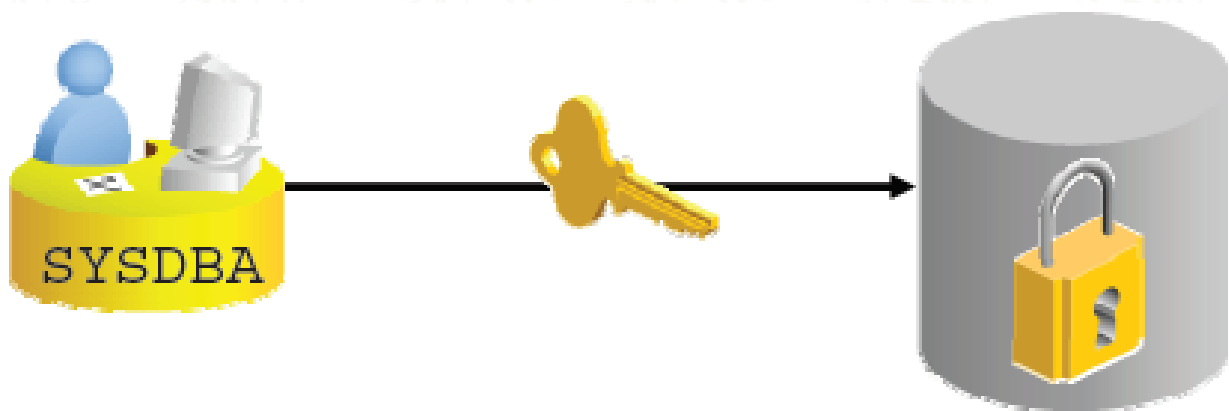
AQ_ADMINISTRATOR_ROLE and MGMT_USER roles. The SYS and SYSTEM accounts are required accounts in the database. They cannot be dropped.

Best practice: Applying the principle of least privilege, these accounts are not used for routine operations. Users who need DBA privileges have separate accounts with the required privileges granted to them.

The SYSBACKUP, SYSDG, and SYSKM users are created to facilitate separation of duties for database administrators. Each of these provides a designated use for an administrative privilege by the same name. You should create a user and grant the appropriate administrative privilege to that user.

Note: Any user that is granted the SYSDBA privilege can connect to the SYS account by using the AS SYSDBA clause.

Note: The SYS and SYSTEM accounts are required accounts in the database. They cannot be dropped.



2.2. Administrative Privileges:

Privilege	Description
SYSDBA	Standard database operations, such as starting and shutting down the database instance, creating the server parameter file (SPFILE), and changing the ARCHIVELOG mode. Allows the grantee to view user data
SYSOPER	Standard database operations, such as starting and shutting down the database instance, creating the server parameter file (SPFILE), and changing the ARCHIVELOG mode
SYSDG	Oracle Recovery Manager (RMAN) backup and recovery operations by using RMAN or SQL*Plus
SYSDG	Data Guard operations by using the Data Guard Broker or the DGMGRL command-line interface
SYSKM	Manage Transparent Data Encryption wallet operations

Note:

Security for SYSDBA, SYSOPER roles is often controlled through access to the OS account owner of the Oracle software. Additionally, security for these roles can be administered via a password file, which allows remote client/server access. Starting with 12c the SYS account could also be locked in some databases. Locking the account prevented unauthorized access from the server and another OS account but SYSDBA will need to be granted to an authorized user first.

2.3. Locking Accounts and Expiring Passwords:

To begin securing your database, you should minimally change the password for every default account and then lock any accounts that you are not using.

Locking an account means that a user won't be able to access it unless a DBA explicitly unlocks it.

Also Consider having policies that change the password for each account. Expiring the password means that when a user first attempts to access an account, that user

will be forced to change the password, but it does not require the current password, so it is better to change passwords and lock the accounts.

After creating a database, it would be better to lock every default account and change their passwords; and to unlock default users only as they are needed.

The following script generates the SQL statements:

```
SQL> alter user <username> identified by <new password>;
SQL> select
'alter user ' || username || ' account lock;'
from dba_users;
```

A locked user can only be accessed by altering the user to an unlocked state; for example:

```
SQL> alter user outln account unlock;
```

A user with an expired password is prompted for a new password when first connecting to the database as that user. When connecting to a user, Oracle checks to see if the current password is expired and, if so, prompts the user, as follows:

```
ORA-28001: the password has expired
Changing password for ...
New password:
```

After entering the new password, the user is prompted to enter it again:

```
Retype new password:
Password changed
Connected.
```

2.4. Checking Default Passwords:

You should also check your database to determine whether any accounts are using default passwords. you can check the `DBA_USERS_WITH_DEFPWD` view whether any Oracle-created user accounts are still set to the default password:

```
SQL> select * from dba_users_with_defpwd;
```

3. Creating Users:

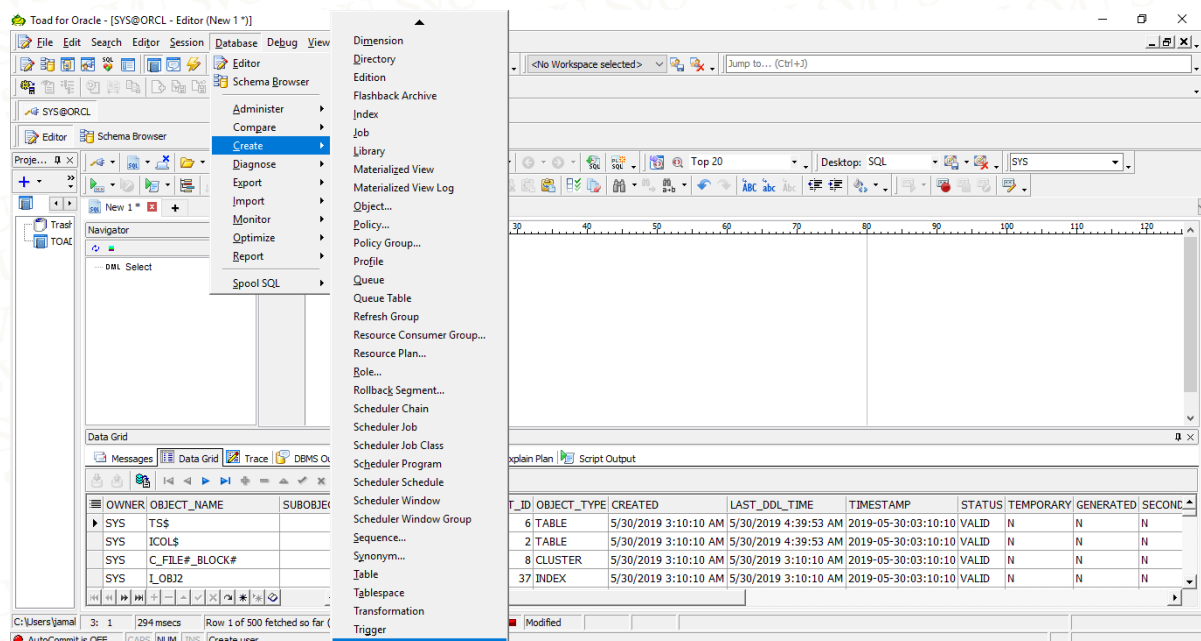
When you are creating a user, you need to consider the following factors:

- Username and authentication method
- Basic privileges
- Default permanent tablespace and space quotas
- Default temporary tablespace

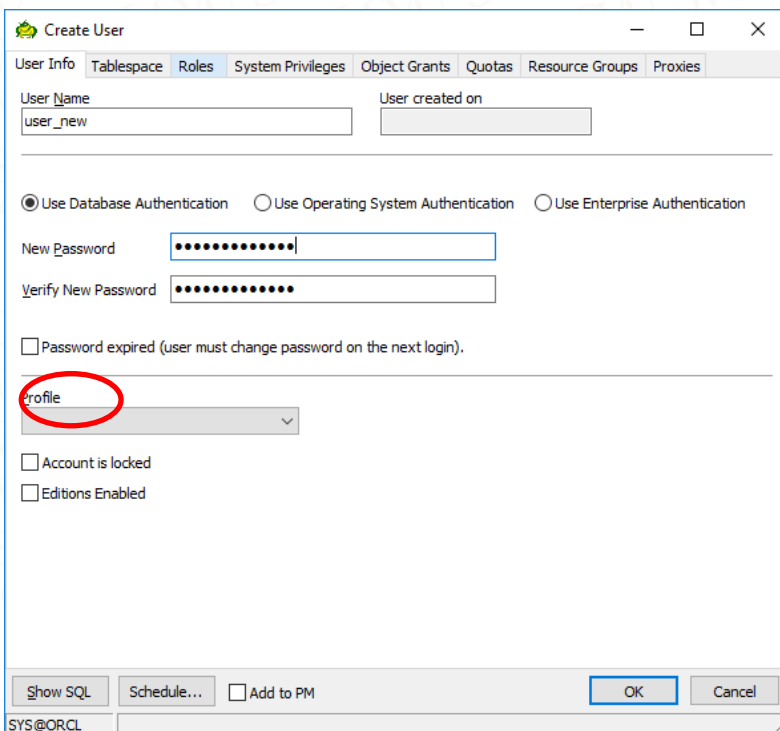
Note: Pluggable database environments have common users and local users. Common users span all pluggable databases within a container database. Local users exist within one pluggable database.

The following figures are show the required steps to create a user account:

The following figure shows the how to choose the create user operation from **Toad** for Oracle application:

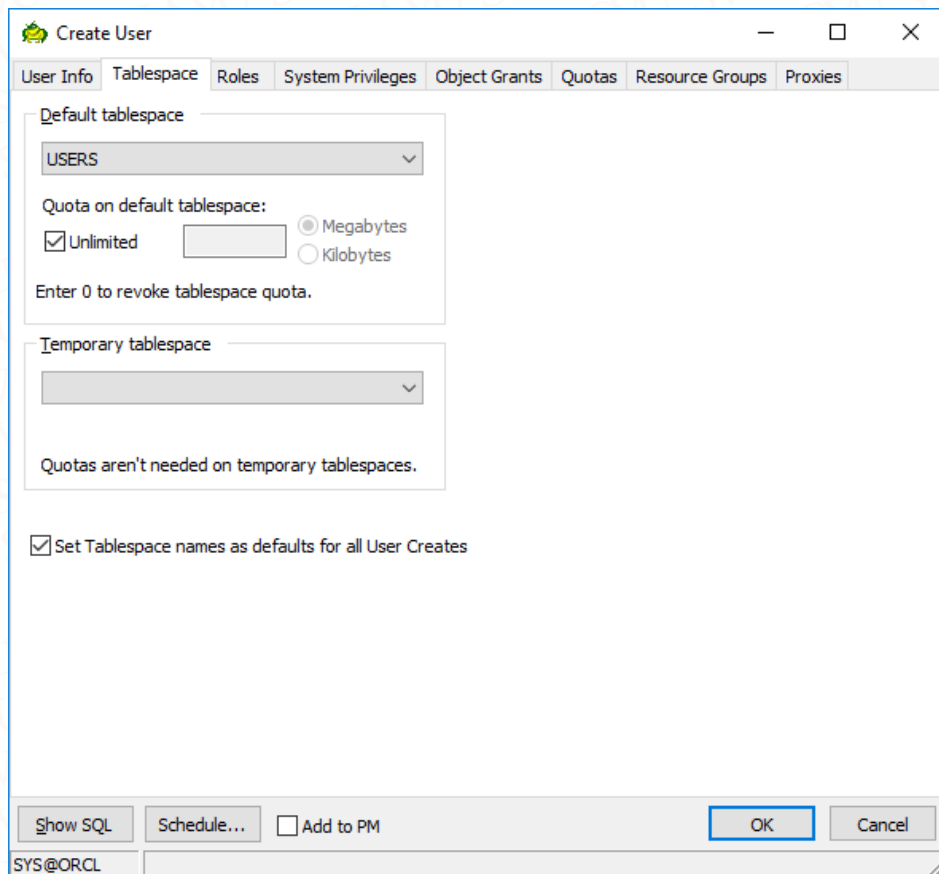


The following figure shows the entering the essential data for the new user; as user name, authentication method, and profile



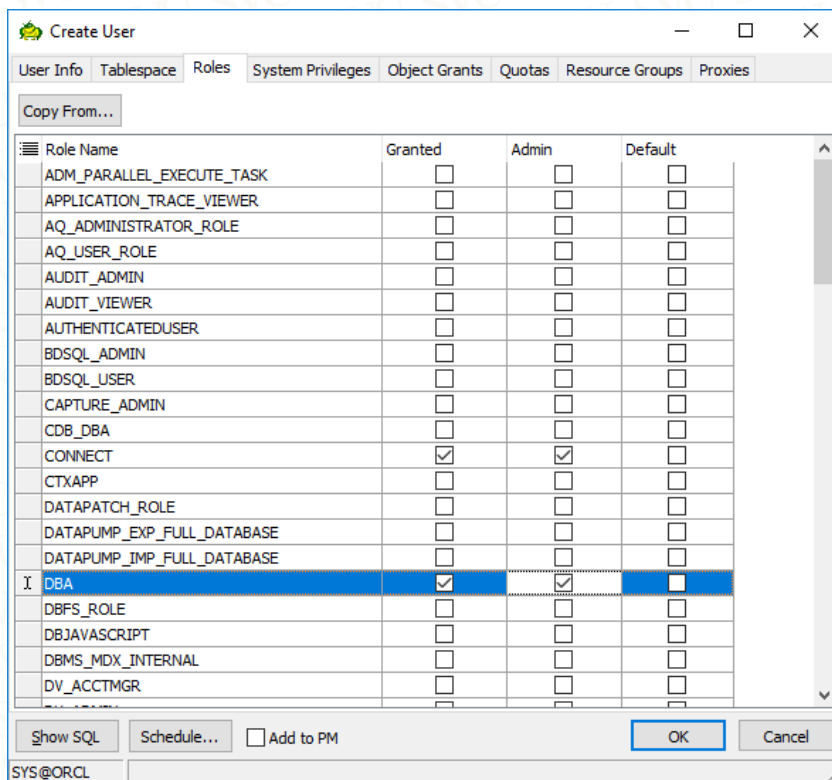
The 'Create User' dialog box is shown with the 'User Info' tab selected. The 'User Name' field contains 'user_new'. The 'User created on' field is empty. The 'Use Database Authentication' radio button is selected. The 'New Password' and 'Verify New Password' fields are filled with masked characters. The 'Password expired' checkbox is unchecked. The 'Profile' dropdown menu is highlighted with a red circle and shows 'DEFAULT' as the selected option. The 'Account is locked' and 'Editions Enabled' checkboxes are unchecked. At the bottom, there are buttons for 'Show SQL', 'Schedule...', 'Add to PM', 'OK', and 'Cancel'. The status bar shows 'SYS@ORCL'.

The following figure shows the assignment of default tablespace for the new user:



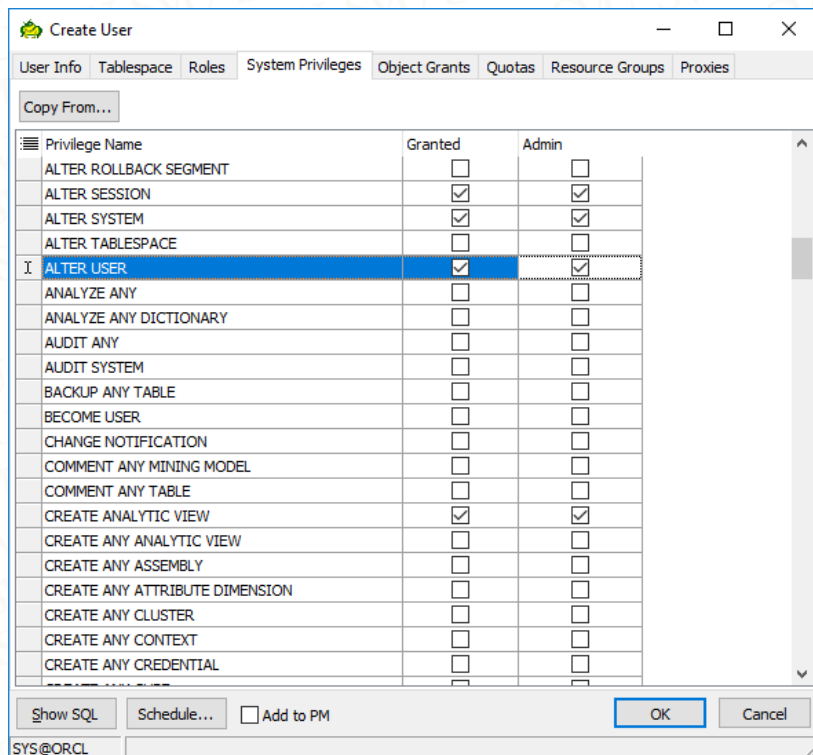
The 'Create User' dialog box is shown with the 'Tablespace' tab selected. The 'Default tablespace' dropdown menu is set to 'USERS'. The 'Quota on default tablespace' section has the 'Unlimited' checkbox checked. The 'Temporary tablespace' dropdown menu is empty. The 'Set Tablespace names as defaults for all User Creates' checkbox is checked. At the bottom, there are buttons for 'Show SQL', 'Schedule...', 'Add to PM', 'OK', and 'Cancel'. The status bar shows 'SYS@ORCL'.

The following figure shows the granting roles for the new user with options:



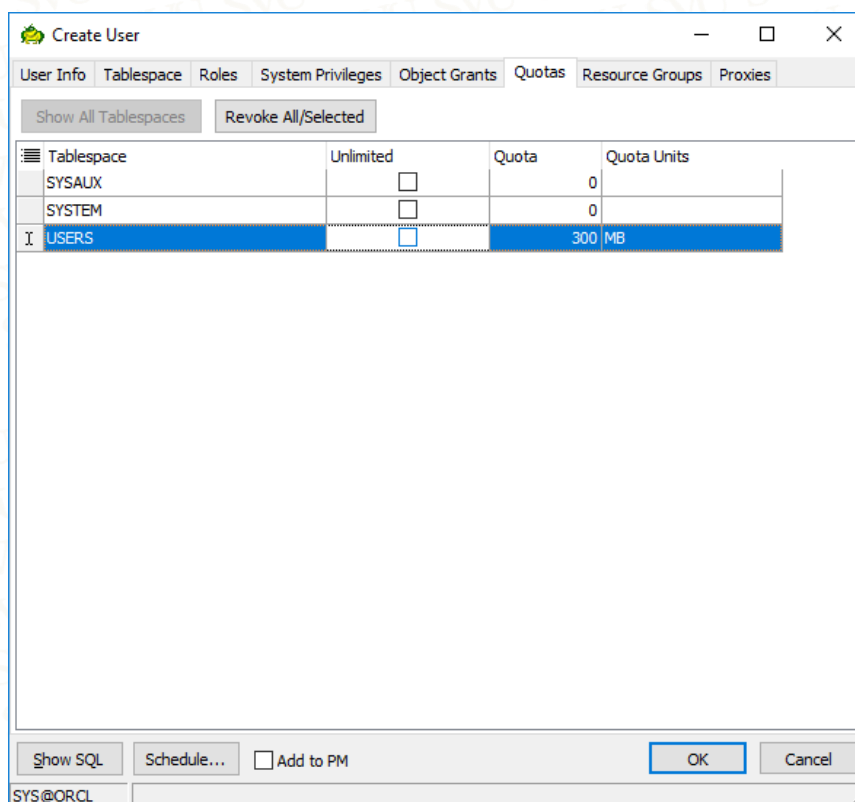
Role Name	Granted	Admin	Default
ADM_PARALLEL_EXECUTE_TASK	☐	☐	☐
APPLICATION_TRACE_VIEWER	☐	☐	☐
AQ_ADMINISTRATOR_ROLE	☐	☐	☐
AQ_USER_ROLE	☐	☐	☐
AUDIT_ADMIN	☐	☐	☐
AUDIT_VIEWER	☐	☐	☐
AUTHENTICATEDUSER	☐	☐	☐
BDSQL_ADMIN	☐	☐	☐
BDSQL_USER	☐	☐	☐
CAPTURE_ADMIN	☐	☐	☐
CDB_DBA	☐	☐	☐
CONNECT	☒	☒	☐
CTXAPP	☐	☐	☐
DATAPATCH_ROLE	☐	☐	☐
DATAPUMP_EXP_FULL_DATABASE	☐	☐	☐
DATAPUMP_IMP_FULL_DATABASE	☐	☐	☐
DBA	☒	☒	☐
DBFS_ROLE	☐	☐	☐
DBJAVASCRIPT	☐	☐	☐
DBMS_MDX_INTERNAL	☐	☐	☐
DV_ACCTMGR	☐	☐	☐

The following figure shows the granting system privileges for the new user:



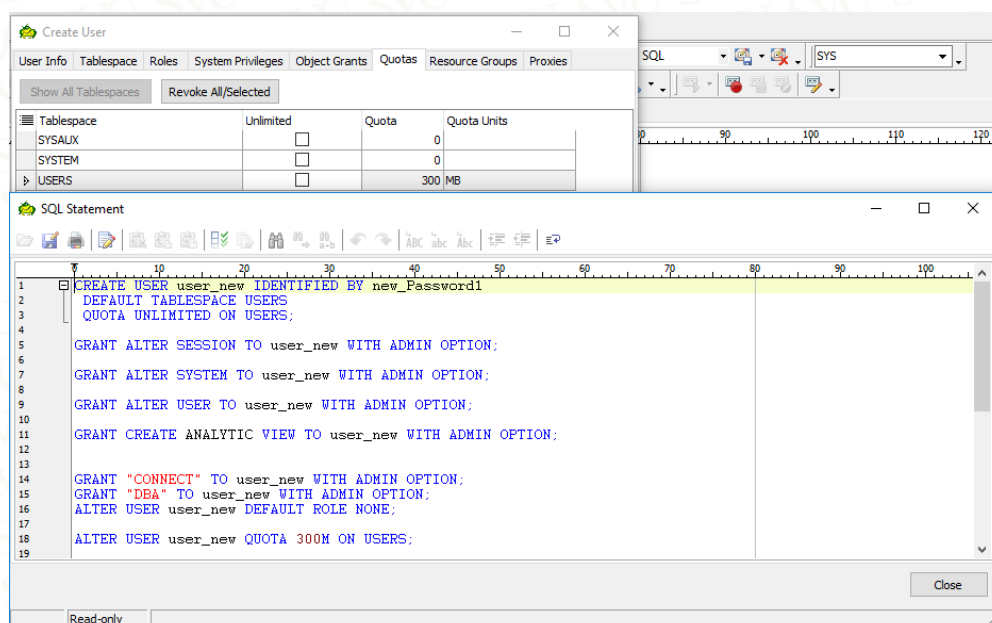
Privilege Name	Granted	Admin
ALTER ROLLBACK SEGMENT	☐	☐
ALTER SESSION	☒	☒
ALTER SYSTEM	☒	☒
ALTER TABLESPACE	☐	☐
ALTER USER	☒	☒
ANALYZE ANY	☐	☐
ANALYZE ANY DICTIONARY	☐	☐
AUDIT ANY	☐	☐
AUDIT SYSTEM	☐	☐
BACKUP ANY TABLE	☐	☐
BECOME USER	☐	☐
CHANGE NOTIFICATION	☐	☐
COMMENT ANY MINING MODEL	☐	☐
COMMENT ANY TABLE	☐	☐
CREATE ANALYTIC VIEW	☒	☒
CREATE ANY ANALYTIC VIEW	☐	☐
CREATE ANY ASSEMBLY	☐	☐
CREATE ANY ATTRIBUTE DIMENSION	☐	☐
CREATE ANY CLUSTER	☐	☐
CREATE ANY CONTEXT	☐	☐
CREATE ANY CREDENTIAL	☐	☐

The following figure shows the quotas' assignment on each existing tablespaces:



Tablespace	Unlimited	Quota	Quota Units
SYSaux	☐	0	
SYSTEM	☐	0	
USERS	☐	300	MB

The following figure is the last step and it shows the create user statement related to previous steps:



```

1 CREATE USER user_new IDENTIFIED BY new_Password1
2   DEFAULT TABLESPACE USERS
3   QUOTA UNLIMITED ON USERS;
4
5 GRANT ALTER SESSION TO user_new WITH ADMIN OPTION;
6
7 GRANT ALTER SYSTEM TO user_new WITH ADMIN OPTION;
8
9 GRANT ALTER USER TO user_new WITH ADMIN OPTION;
10
11 GRANT CREATE ANALYTIC VIEW TO user_new WITH ADMIN OPTION;
12
13
14 GRANT "CONNECT" TO user_new WITH ADMIN OPTION;
15 GRANT "DBA" TO user_new WITH ADMIN OPTION;
16 ALTER USER user_new DEFAULT ROLE NONE;
17
18 ALTER USER user_new QUOTA 300M ON USERS;
19

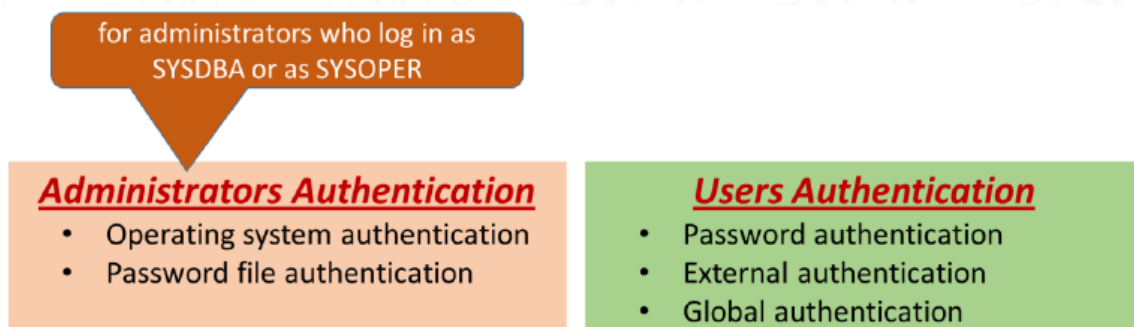
```

3.1. Authentication Methods:

A user account must have an authentication method: some means whereby the database can determine if the user attempting to create a session connecting to the account is allowed to do so. Authentication types are:

- Operating system authentication (for administrators)
- Password file authentication (for administrators)
- Password authentication (for users)
- External authentication (for users)
- Global authentication (for users)

The first two techniques are used only for administrators who log in as SYSDBA or as SYSOPER; the last requires an LDAP directory server. The LDAP directory server is the Oracle Internet Directory, shipped as a part of the Oracle Application Server.



Operating System Authentication:

- Type of Administrators Authentication who log in as SYSDBA or as SYSOPER
- Users log on without specifying additional password or user name
- Best when only local access or secured line access are allowed
- Can be a security risk if used with non-secure internet or network lines
- Requires knowledge of operating system security
- Two special operating system groups control database administrator connections when using OS authentication
- These groups are generically referred to as OSDBA and OSOPER

How to implement

1. Create operating system user for DBA use
2. Create OSDBA group (UNIX only because Windows automatically creates the group with name ORA_DBA)
3. Assign operating system user to ORA_DBA group in Windows or OSDBA group in Unix:

```
Set REMOTE_LOGIN_PASSWORDFILE = NONE  
CONNECT / @oradb as sysdba
```

Password File Authentication:

- Type of Administrators Authentication who log in as SYSDBA or as SYSOPER
- Following database creation, the only user with these privileges is SYS
- Users must know their Oracle username and password to log onto the database
- Best for non-secured line access, such as Internet, is allowed
- Users required to remember multiple usernames and passwords
- Possible security risk due to storage of passwords in an encrypted operating system file

How to implement:

1. Set REMOTE_LOGIN_PASSWORDFILE = EXCLUSIVE (or SHARED if database is part of distributed system)
2. Add more SYSDBA or SYSOPER users by Creating new Oracle user, for example student:

```
CREATE USER student IDENTIFIED BY mypass;
```
3. Assigning SYSDBA or SYSOPER privileges to this user:

```
GRANT sysdba TO student;
```

Password file is automatically updated when new users are assigned these privileges and when any of these users change their passwords:

```
SELECT * FROM v$pwfile_users;  
CONNECT student/mypass as sysdba;
```

Note: OS authentication takes precedence over password file authentication for privileged users

Password Authentication:

1. Authentication by the Oracle database
2. Encrypted and stored in the data dictionary using the Advanced Encryption Standard (AES) algorithm during network connections before sending them across the network.
3. When setting up a password, you can expire the password immediately, which forces the user to change the password after first logging in

External Authentication:

1. Authentication is delegated to an external service
2. Users can connect without user name and password
3. Operating System

```
OS_AUTHENT_PREFIX = OPS$  
Windows account: John  
Windows domain: SVU  
Create user "OPS$SVU\JOHN" identified externally;  
Connect /
```

Global Authentication:

1. Requires the Oracle Advanced Security option
2. Enables users to be identified through the use of Oracle Internet Directory

3.2. Grant Privileges:

By default, no one can do anything in an Oracle database. A user cannot even connect without being granted a privilege. And once this has been done, he still can't do anything useful (or dangerous) without being given more privileges. Privileges are assigned to user accounts with a `GRANT` command and withdrawn with a `REVOKE`. By

default, only the DBAs (`SYS` and `SYSTEM`) have the right to grant any but the most limited privileges.

There are two types of user privileges:

System Privileges:

There are about two hundred system privileges. Most apply to actions that affect the data dictionary, such as creating tables or users. Others affect the database or the instance, such as creating tablespaces, adjusting instance parameter values, or establishing a session.

- Some of the more commonly used privileges are: `CREATE SESSION`, `CREATE TABLE`, `CREATE ANY TABLE`, `SELECT ANY TABLE`
- `ANY` clause means that the privilege crosses schema lines:
`Grant select any table to scott with admin option;`
- The `SYS` user and users with the `DBA` role are granted all of the `ANY` privileges

Object Privileges:

Object privileges give the ability to perform `SELECT`, `INSERT`, `UPDATE`, and `DELETE` commands against tables and related objects, and to execute PL/SQL objects. These privileges do not exist for objects in the users' own schemas; if a user has the system privilege `CREATE TABLE`, he/she can perform `SELECT` and DML operations against the tables he/she creates with no further need for permissions.

The object privileges apply to different types of object:

Privilege	Granted on
<code>SELECT</code>	Tables, views, sequences, synonyms
<code>INSERT</code>	Tables, views, synonyms
<code>UPDATE</code>	Tables, views, synonyms
<code>DELETE</code>	Tables, views, synonyms
<code>ALTER</code>	Tables, sequences
<code>EXECUTE</code>	Procedures, functions, packages, synonyms

Using
WITH

`GRANT OPTION`, lets a user pass his/her object privilege on to a third party:
`Grant select on employees to scott with grant option;`

3.3. Revoking Privileges:

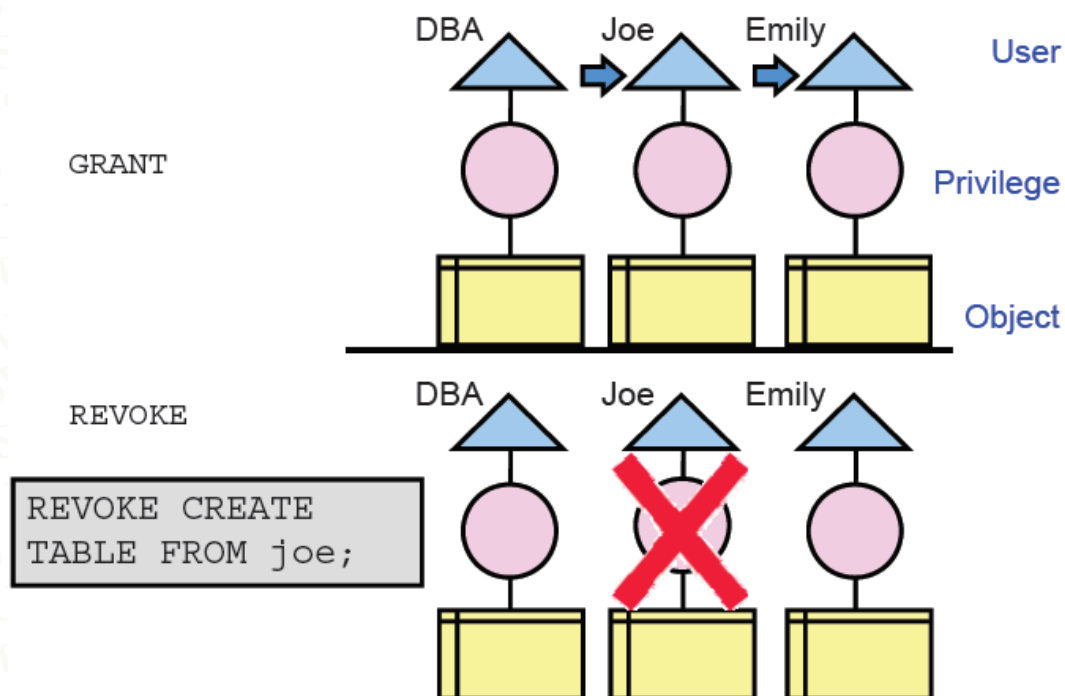
Revoking System Privileges with ADMIN OPTION

System privileges that have been granted directly with a GRANT command can be revoked by using the REVOKE SQL statement. Users with ADMIN OPTION for a system privilege can revoke the privilege from any other database user. The revoker does not have to be the same user who originally granted the privilege. There are no cascading effects when a system privilege is revoked, regardless of whether it is given the ADMIN OPTION.

The SQL syntax for revoking system privileges is:

```
REVOKE <system_privilege> FROM <grantee clause>
```

The enclosed figure illustrates the following situation.



Scenario:

1. The DBA grants the CREATE TABLE system privilege to Joe with ADMIN OPTION.
2. Joe creates a table.
3. Joe grants the CREATE TABLE system privilege to Emily.
4. Emily creates a table.
5. The DBA revokes the CREATE TABLE system privilege from Joe.

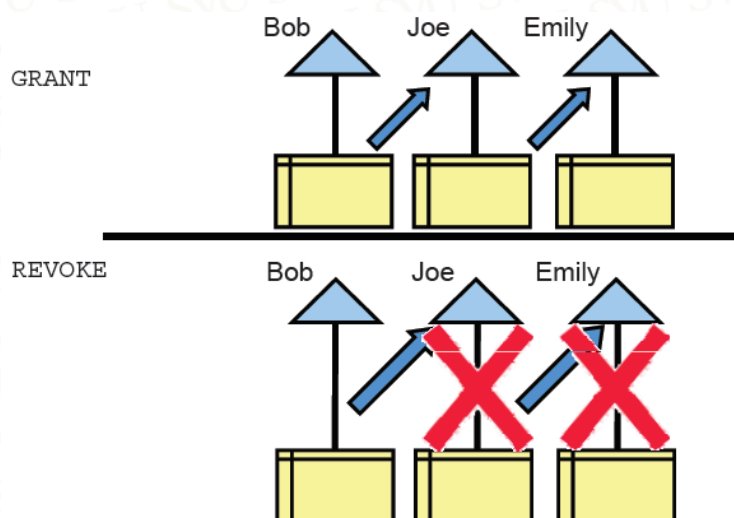
Result:

Joe's table still exists, but Joe cannot create new tables. Emily's table still exists, and she still has the CREATE TABLE system privilege.

Revoking Object Privileges with GRANT OPTION

Cascading effects can be observed when revoking a system privilege that is related to a data manipulation language (DML) operation. For example, if the SELECT ANY TABLE privilege is granted to a user, and if that user has created procedures that use the table, all procedures that are contained in the user's schema must be recompiled before they can be used again.

Revoking object privileges also cascades when given with GRANT OPTION. As a user, you can revoke only those privileges that you have granted. For example, Bob cannot revoke the object privilege that Joe granted to Emily. Only the grantee or a user with the privilege called GRANT ANY OBJECT PRIVILEGE can revoke object privileges.



Scenario:

1. Joe is granted the `SELECT` object privilege on `EMPLOYEES` with `GRANT OPTION`.
2. Joe grants the `SELECT` privilege on `EMPLOYEES` to Emily.
3. The `SELECT` privilege is revoked from Joe. This revoke is cascaded to Emily as well.

3.4. Using Roles to Manage Privileges:

A role is a named group of related privileges that are granted to users or to other roles. You can use roles to administer database privileges. You can add privileges to a role and grant the role to a user. The user can then enable the role and exercise the privileges granted by the role. A role contains all privileges that are granted to that role and all privileges of other roles that are granted to it.

There are several roles defined automatically for Oracle databases when you run database creation scripts.

Please check this example to learn how to manage roles and this video to see how to create roles in Toad for Oracle application

Roles provide the following benefits with respect to managing privileges:

Easier privilege management:

Use roles to simplify privilege management. Rather than granting the same set of privileges to several users, you can grant the privileges to a role and then grant that role to each user.

- Privileges are granted to and revoked from roles as though the role were a user.
- Roles are granted to and revoked from users or other roles as though they were system privileges.
- A role can consist of both system and object privileges.
- A role can be enabled or disabled for each user who is granted the role.
- A role can require a password to be enabled.
- Roles are not owned by anyone, and they are not in any schema.

Dynamic privilege management:

If the privileges associated with a role are modified, all users who are granted the role acquire the modified privileges automatically and immediately.

Selective availability of privileges:

Roles can be enabled and disabled to turn privileges on and off temporarily. This allows the privileges of the user to be controlled in a given situation.

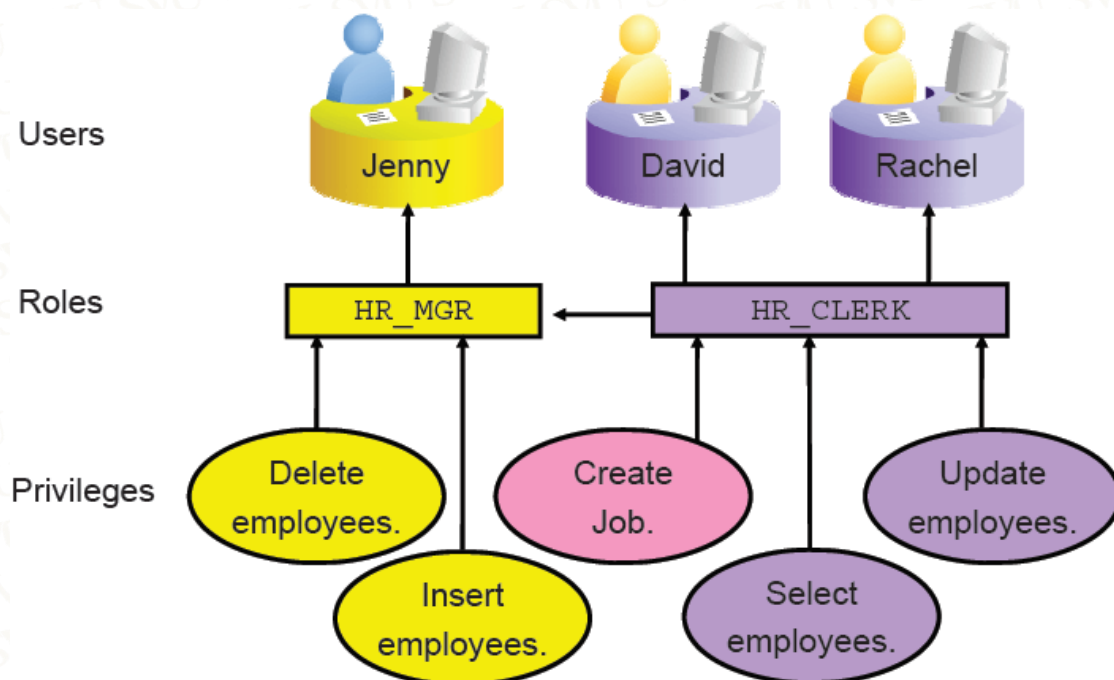
Predefined Roles

Role	Privileges Included
CONNECT	CREATE SESSION
DBA	Most system privileges; several other roles. Do not grant to non-administrators.
RESOURCE	CREATE CLUSTER, CREATE INDEXTYPE, CREATE OPERATOR, CREATE PROCEDURE, CREATE SEQUENCE, CREATE TABLE, CREATE TRIGGER, CREATE TYPE
SCHEDULER_ADMIN	CREATE ANY JOB, CREATE EXTERNAL JOB, CREATE JOB, EXECUTE ANY CLASS, EXECUTE ANY PROGRAM, MANAGE SCHEDULER
SELECT_CATALOG_ROLE	SELECT privileges on data dictionary objects

Example:

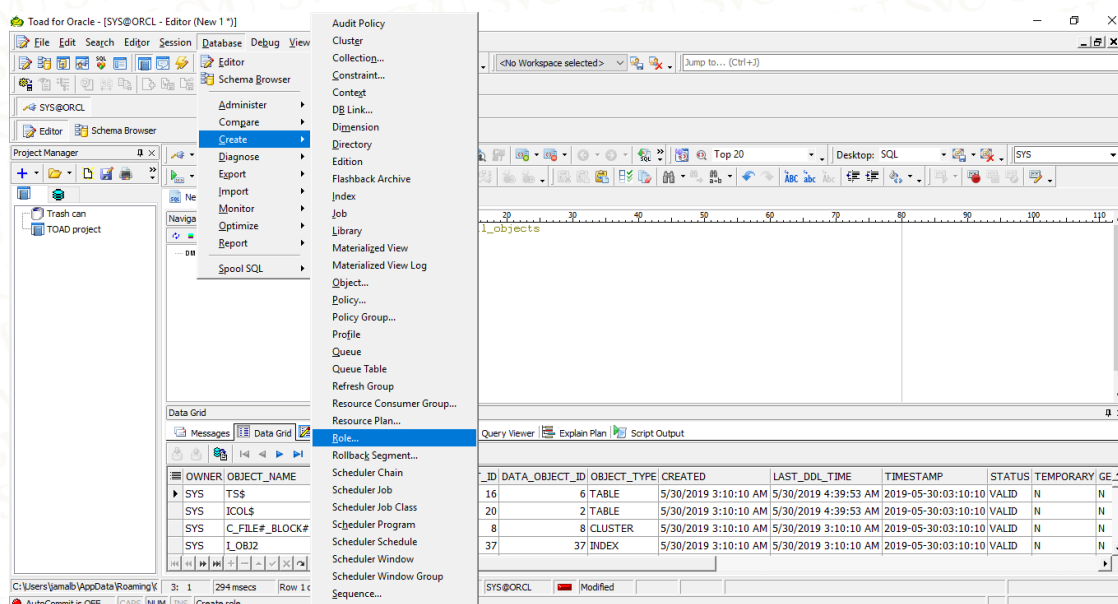
In the following example, the SELECT and UPDATE privileges on the employees table and the CREATE JOB system privilege are granted to the HR_CLERK role. DELETE and INSERT privileges on the employees table and the HR_CLERK role are granted to the HR_MGR role.

The manager is granted the HR_MGR role and can now select, delete, insert, and update the employees table.



The following figures are shown the steps to create a role:

The following figure shows the how to choose the create new role operation from Toad for Oracle application



The following figure shows the create new Role Info:

Create Role

Role Info | Roles | System Privileges | Object Grants | Resource Groups

Role Name
role_new

☒ Not Identified
☐ Use Database Authentication
☐ Use Operating System Authentication
☐ Use Enterprise Authentication

New Password

Verify New Password

Show SQL | Schedule... | ☐ Add to PM | OK | Cancel

SYS@ORCL

The following figure shows how to grant roles to the new Role:

Create Role

Role Info | Roles | System Privileges | Object Grants | Resource Groups

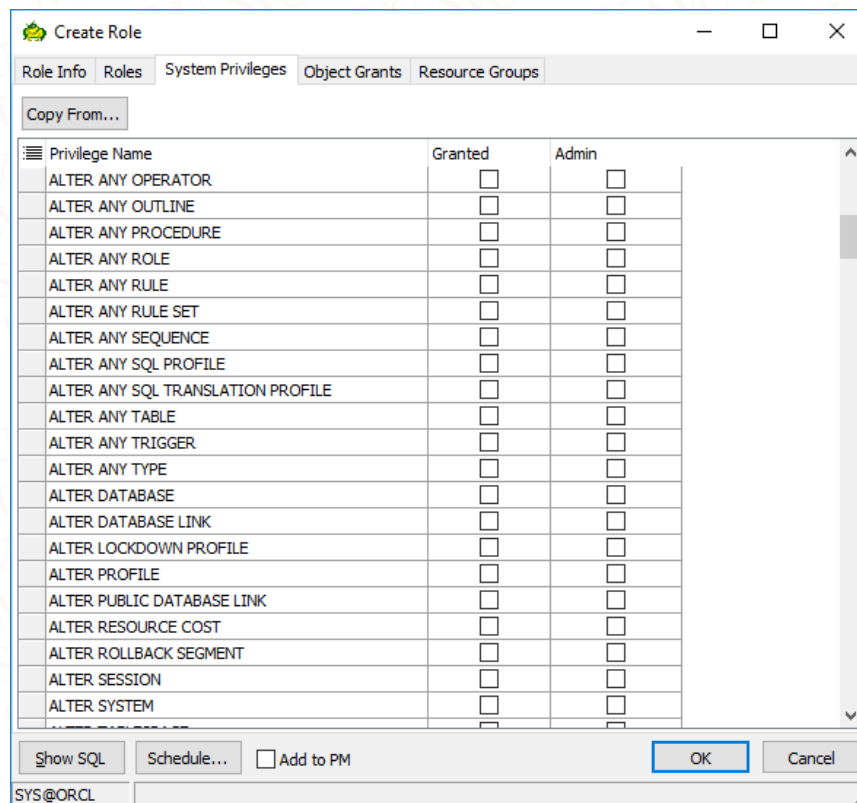
Copy From...

Role Name	Granted	Admin
ADM_PARALLEL_EXECUTE_TASK	☐	☐
APPLICATION_TRACE_VIEWER	☐	☐
AQ_ADMINISTRATOR_ROLE	☐	☐
AQ_USER_ROLE	☐	☐
AUDIT_ADMIN	☐	☐
AUDIT_VIEWER	☐	☐
AUTHENTICATEDUSER	☐	☐
BDSQL_ADMIN	☐	☐
BDSQL_USER	☐	☐
CAPTURE_ADMIN	☐	☐
CDB_DBA	☐	☐
CONNECT	☒	☒
CTXAPP	☐	☐
DATAPATCH_ROLE	☐	☐
DATAPUMP_EXP_FULL_DATABASE	☐	☐
DATAPUMP_IMP_FULL_DATABASE	☐	☐
DBA	☒	☒
DBFS_ROLE	☐	☐
DBJAVASCRIPT	☐	☐
DBMS_MDX_INTERNAL	☐	☐
DV_ACCTMGR	☐	☐

Show SQL | Schedule... | ☐ Add to PM | OK | Cancel

SYS@ORCL

The following figure shows how to grant System Privileges to the new Role:



Create Role

Role Info | Roles | **System Privileges** | Object Grants | Resource Groups

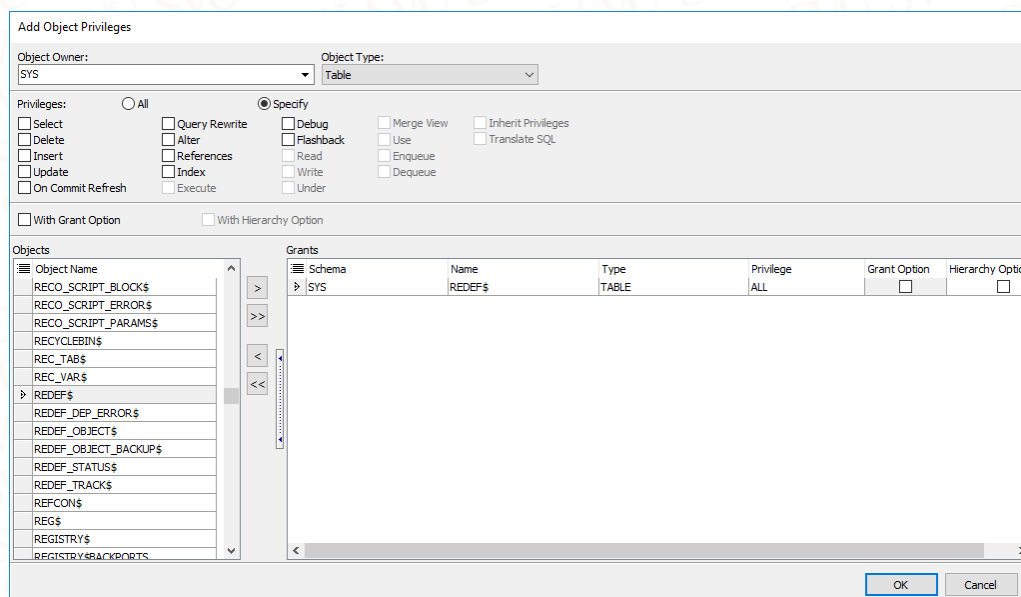
Copy From...

Privilege Name	Granted	Admin
ALTER ANY OPERATOR	☐	☐
ALTER ANY OUTLINE	☐	☐
ALTER ANY PROCEDURE	☐	☐
ALTER ANY ROLE	☐	☐
ALTER ANY RULE	☐	☐
ALTER ANY RULE SET	☐	☐
ALTER ANY SEQUENCE	☐	☐
ALTER ANY SQL PROFILE	☐	☐
ALTER ANY SQL TRANSLATION PROFILE	☐	☐
ALTER ANY TABLE	☐	☐
ALTER ANY TRIGGER	☐	☐
ALTER ANY TYPE	☐	☐
ALTER DATABASE	☐	☐
ALTER DATABASE LINK	☐	☐
ALTER LOCKDOWN PROFILE	☐	☐
ALTER PROFILE	☐	☐
ALTER PUBLIC DATABASE LINK	☐	☐
ALTER RESOURCE COST	☐	☐
ALTER ROLLBACK SEGMENT	☐	☐
ALTER SESSION	☐	☐
ALTER SYSTEM	☐	☐

☐ Add to PM

SYS@ORCL

The following figure shows how to grant Object Privileges to the new Role:



Add Object Privileges

Object Owner: SYS Object Type: Table

Privileges: ☐ All ☒ Specify

☐ Select ☐ Query Rewrite ☐ Debug ☐ Merge View ☐ Inherit Privileges
☐ Delete ☐ Alter ☐ Flashback ☐ Use ☐ Translate SQL
☐ Insert ☐ References ☐ Read ☐ Enqueue
☐ Update ☐ Index ☐ Write ☐ Dequeue
☐ On Commit Refresh ☐ Execute ☐ Under

☐ With Grant Option ☐ With Hierarchy Option

Objects

Object Name	Name	Type	Privilege	Grant Option	Hierarchy Option
RECO_SCRIPT_BLOCK\$	REDEF\$	TABLE	ALL	☐	☐
RECO_SCRIPT_ERROR\$					
RECO_SCRIPT_PARAMS\$					
RECYCLEBIN\$					
REC_TAB\$					
REC_VAR\$					
REDEF\$					
REDEF_DEP_ERROR\$					
REDEF_OBJECT\$					
REDEF_OBJECT_BACKUP\$					
REDEF_STATUS\$					
REDEF_TRACKS\$					
REPCON\$					
REG\$					
REGISTRY\$					
REGISTRYBACKPORTS					

The following figure shows how to grant Object Privileges to the new Role:

The following figure shows the SQL statement generated with the previous information:

```

1 CREATE ROLE role_new IDENTIFIED GLOBALLY;
2
3 GRANT ALTER SESSION TO role_new WITH ADMIN OPTION;
4
5 GRANT ALTER SYSTEM TO role_new WITH ADMIN OPTION;
6
7 GRANT 'CONNECT' TO role_new WITH ADMIN OPTION;
8
9 GRANT 'DBA' TO role_new WITH ADMIN OPTION;
10
11 GRANT ALL ON 'SYS'. 'REDEF$' TO role_new;
12

```

3.5. Managing Profiles:

Users are assigned only one profile at a time.

Profiles:

- Control resource consumption
- Manage account status and password expiration

Profiles impose a named set of resource limits on database usage and instance resources. Profiles also manage the account status and place limitations on users' passwords (length, expiration time, and so on). Every user is assigned a profile and may belong to only one profile at any given time. If users have already logged in when you change their profile, the change does not take effect until their next login.

The `DEFAULT` profile serves as the basis for all other profiles (applied by default to all users, including `SYS` and `SYSTEM`)

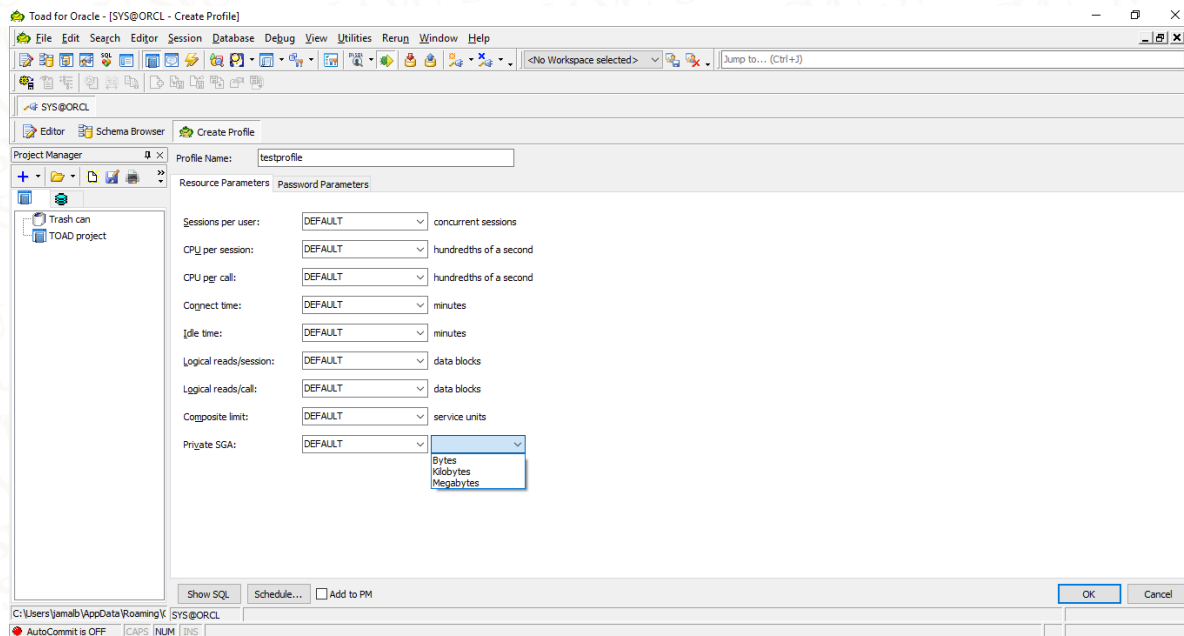
```
select profile, resource_name, resource_type, limit
from dba_profiles
order by profile, resource_type;
```

Unlike password settings, kernel resource profile restrictions don't take effect until you set the `RESOURCE_LIMIT` initialization parameter to `TRUE` for your database; for example:

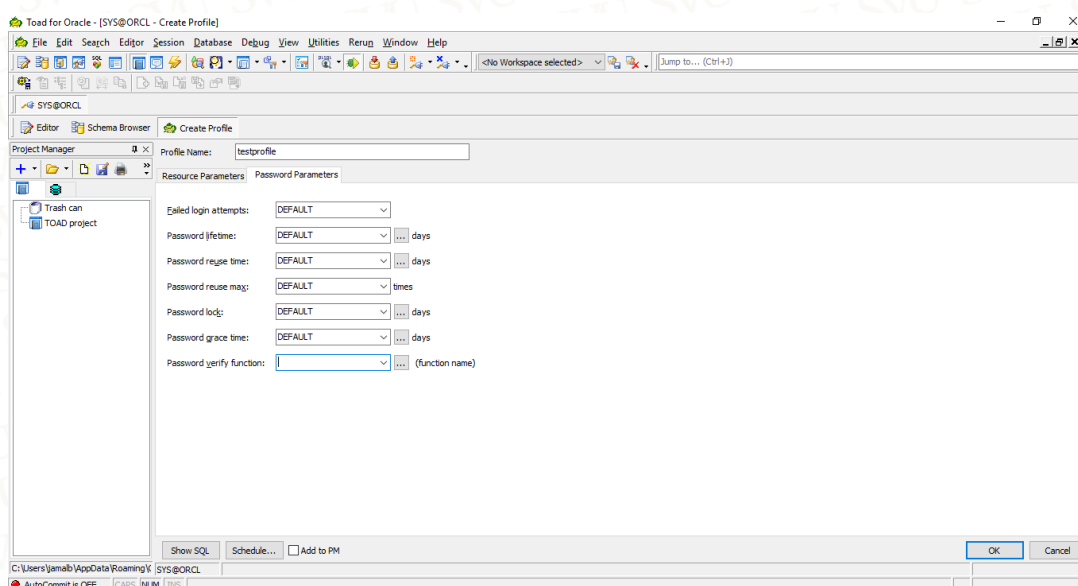
```
SQL> alter system set resource_limit=true scope=both;
SQL> select name, value from v$parameter where
name='resource_limit';
```

The following figures are shown the steps to create a profile:

The following figure shows the how to choose the create new profile operation from Toad for Oracle application –first page information: Resource Parameters:



The following figure shows the how to choose the create new profile operation from Toad for Oracle application –second page information: Password Parameters:



3.6. Implementing Password Security Features:

Oracle password management is implemented with user profiles. Profiles can provide many standard security features:



Account locking:

Enables automatic locking of accounts for a set duration when users fail to log in to the system in the specified number of attempts

- `FAILED_LOGIN_ATTEMPTS`: Specifies the number of failed login attempts before the lockout of the account
- `PASSWORD_LOCK_TIME`: Specifies the number of days for which the account is locked after the specified number of failed login attempts

Note: Do not use profiles that cause the `SYS`, `SYSMAN`, and `DBSNMP` passwords to expire and the accounts to be locked.

Password aging and expiration:

Enables user passwords to have a lifetime, after which the passwords expire and must be changed

- **PASSWORD_LIFE_TIME:** Determines the lifetime of the password in days, after which the password expires
- **PASSWORD_GRACE_TIME:** Specifies a grace period in days for changing the password after the first successful login after the password has expired

Password history:

Checks the new password to ensure that the password is not reused for a specified amount of time or a specified number of password changes. These checks can be implemented by using one of the following:

- **PASSWORD_REUSE_TIME:** Specifies that a user cannot reuse a password for a given number of days
- **PASSWORD_REUSE_MAX:** Specifies the number of password changes that are required before the current password can be reused

Password complexity verification:

Makes a complexity check on the password to verify that it meets certain rules. The check must ensure that the password is complex enough to provide protection against intruders who may try to break into the system by guessing the password.

It is useful to check to see whether the password for a user has ever been changed:

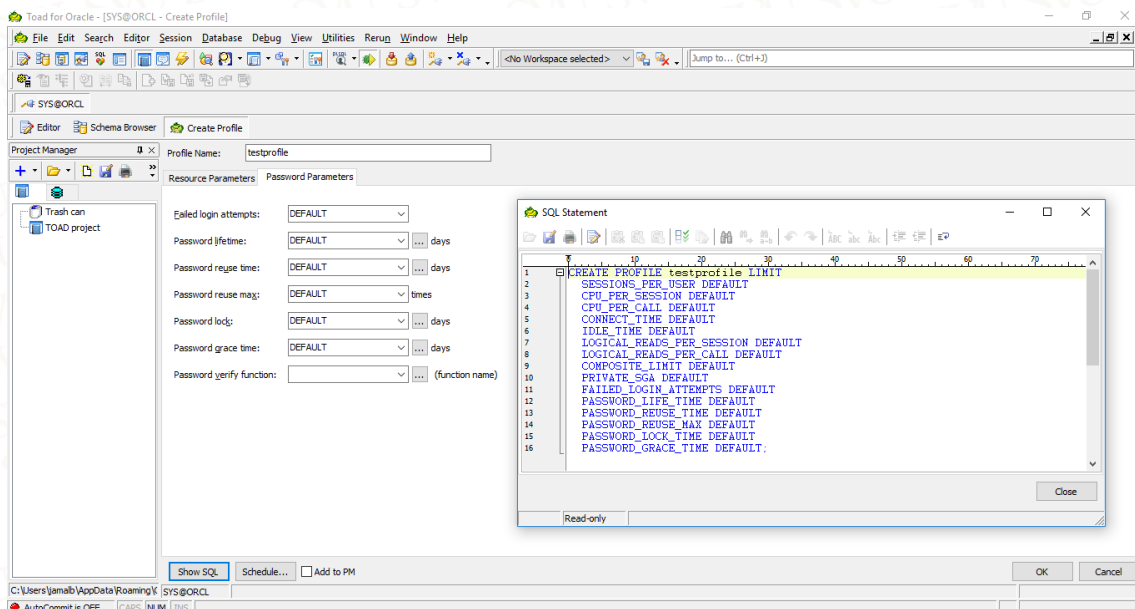
```
select
name user_name, to_char(ctime,'dd-mon-yy hh24:mi:ss')
password_creation_time
,to_char(pptime,'dd-mon-yy hh24:mi:ss') password_present_time
,length(password)
from user$
where password is not null
and password not in ('GLOBAL','EXTERNAL')
and ctime=pptime;
```

3.7. Supplied Password Verification Functions:

- The following functions are created by the
\$ORACLE_HOME/rdbms/admin/utlpwdmg.sql script:
 - ORA12C_VERIFY_FUNCTION
 - ORA12C_STRONG_VERIFY_FUNCTION
- The functions require the following of passwords:
 - Have a minimum number of characters
 - Not be the username, username with a number, or username reversed
 - Not be the database name or the database name with a number
 - Have at least one alphabetic and one numeric character
 - Differ from the previous password by at least three letters

Note: The functions must be owned by the SYS user. Password complexity checking is not enforced for the SYS user.

The following figure shows the how to choose the create new profile operation from Toad for Oracle application –SQL statement to create the new profile with the selected parameters:



4. Modifying Passwords:

Use the `ALTER USER` command to modify an existing user's password. This example changes the `HR` user's password to `PEOPLE`:

```
SQL> alter user HR identified by PEOPLE;
```

You can change the password of another account only if you have the `ALTER USER` privilege granted to your user. This privilege is granted to the `DBA` role. After you change a password for a user, any subsequent connection to the database by that user requires the password indicated by the `ALTER USER` statement.

When you modify a password, it is case sensitive. The behavior with the case is set with a parameter `SEC_CASE_SENSITIVE_LOGON` and by default is `TRUE`.

You can change the password for a user with the `SQL*Plus PASSWORD` command. After issuing the command, you are prompted for a new password:

```
SQL> passw HR
Changing password for HR
New password:
Retype new password:
Password changed
```

This method has the advantage of changing a password for a user without displaying the new password on the screen.

5. Schema Only Account:

Previously without a schema only account, a DBA would have to log in as a different user to perform some create table scripts and issue grants. Now, there are new Schema Only Accounts that can be created without passwords. This is for application schemas and holds all of the objects, so changes to these objects can be done if granted the privilege to access these accounts. These accounts have no privilege to log in directly to the database. So even without a password, there is no possibility to log in, and an error will occur.

```
SQL> create user appl NO AUTHENTICATION;
```

In the `dba_users` table, this user will have an `AUTHENTICATION_TYPE=NONE`, and the password column will also be `NULL`. Schema only accounts can have privileges granted to create table and other objects; however, it cannot have any of the administrative privileges assigned to it. Even granting `CONNECT SESSION` to the schema only account will not allow you to log in to this schema account directly.

```
SQL> grant create session to appl;
```

Grant succeeded.

```
SQL> connect appl
```

Enter password:

ERROR:

ORA-01005: null password given; logon denied

Warning: You are no longer connected to ORACLE.

```
SQL> connect appl
```

Enter password:

ERROR:

ORA-01017: invalid username/password; logon denied

In order to perform the DDL statements in other accounts including the schema only accounts, a proxy connection can be made. Using a proxy connection was possible, and with the schema only account, it continues to be possible in order to perform any

necessary code as the schema. Here is an example using `jb_dba` as our account as we log in to the database, and the schema only account is `app1`.

```
SQL> alter user app1 grant connect through jb_dba;
SQL> connect jb_dba/password1
SQL>      select      sys_context('USERENV','SESSION_USER')      as
session_user,
      sys_context('USERENV','SESSION_SCHEMA') as session_schema,
      sys_context('USERENV','PROXY_USER') as proxy_id,
      user
      from dual;
```

SESSION_USER	SESSION_SCHEMA	PROXY_ID	USER
-----	-----	-----	-----
APP1	APP1	JB_DBA	APP1

This schema without a password can simply be used to application schemas and allow privileges to be granted to the objects or become this schema to run code. This is something to consider for application schemas; and, as we will see in grants and permissions can still be handled by roles for these objects.

6. Modifying/Dropping Users:

6.1. Modifying users:

Sometimes you need to modify existing users for the following types of reasons:

- Change a user's password
- Lock or unlock a user
- Change the default permanent or temporary tablespace, or both
- Change a profile or role
- Change system or object privileges
- Modify quotas on tablespaces

Use the ALTER USER statement to modify users.

Listed next are several SQL statements that modify a user:

This example changes a user's password, using the IDENTIFIED BY clause:

```
SQL> alter user HR identified by PEOPLE_New12;
```

If you don't set a default permanent tablespace and temporary tablespace when you initially create the user, you can modify them after creation, as shown here:

```
SQL> alter user HR default tablespace users temporary tablespace temp;
```

This example locks a user account:

```
SQL> alter user HR account lock;
```

And, this example alters the user's quota on the USERS tablespace:

```
SQL> alter user HR quota 500m on users;
```

6.2. Dropping Users:

Before you drop a user, it is recommended that you first lock the user. Locking the user prevents others from connecting to a locked database account. This allows you to better determine whether someone is using the account before it is dropped.

Locking users is a very handy technique for securing your database and discovering which users are active. Be aware that by locking a user, you are not locking access to a user's objects. For instance, if a `USER_A` has select, insert, update, and delete privileges on tables owned by `USER_B`, if you lock the `USER_B` account, `USER_A` can still issue DML statements against the objects owned by `USER_B`.

After you are sure that a user and its objects are not needed, use the **DROP USER** statement to remove a database account.

Here is an example of locking a user:

```
SQL> alter user HR account lock;
```

Any user or application attempting to connect to this user now receives the following error:

```
ORA-28000: the account is locked
```

To view the users and lock dates in your database, issue this query:

```
SQL> select username, lock_date from dba_users;
```

To unlock an account, issue this command:

```
SQL> alter user HR account unlock;
```

This example drops the user `HR`:

```
SQL> drop user HR;
```

The prior command won't work if the user owns any database objects. Use the **CASCADE** clause to remove a user and have its objects dropped:

```
SQL> drop user HR cascade;
```

7. Quiz:

1. If you create a user without specifying a temporary tablespace, what temporary tablespace will be assigned? (Choose the best answer).
 - a) You must specify a temporary tablespace
 - b) SYSTEM
 - c) TEMP
 - d) The database default temporary tablespace
 - e) He/she will not have a temporary tablespace
2. Which of these statements about system privileges are correct? (Choose all correct answers).
 - a) Only the SYS and SYSTEM users can grant system privileges.
 - b) If a system privilege is revoked from a user, it will also be revoked from all users to whom he granted it.
 - c) If a system privilege is revoked from a user, it will not be revoked from all users to whom she granted it.
 - d) CREATE TABLE is a system privilege.
 - e) CREATE ANY TABLE is a system privilege.
3. Which of these statements is incorrect regarding roles? (Choose the best answer).
 - a) You can grant object privileges and system privileges and roles to a role.
 - b) A role cannot have the same name as a table.
 - c) A role cannot have the same name as a user.
 - d) Roles can be enabled or disabled within a session.

4. Which of these can be controlled by a password profile? (Choose all correct answers).
- a) Two or more users choosing the same password
 - b) Preventing the reuse of a password by the same user
 - c) Forcing a user to change password
 - d) Enabling or disabling password file authentication
5. All passwords created in Oracle Database are not case-sensitive by default.
- a) True
 - b) False
6. A database role: (Choose all correct answers).
- a) Can be enabled or disabled
 - b) Can consist of system and object privileges
 - c) Is owned by its creator
 - d) Cannot be protected by a password
7. With RESOURCE_LIMIT set at its default value of FALSE, profile password limitations are ignored.
- a) True
 - b) False
8. A database user account is:
- a) A named set of resource limits on database usage
 - b) An authentication by the Oracle database.
 - c) A right to execute a particular type of SQL statement or to access another user's object.
 - d) A named group of related privileges that are granted to users or to other roles.
 - e) A way to organize the ownership of and access to database objects.

9. Users can access only accounts with status?

- a) Locked and Expired
- b) Locked
- c) Open
- d) Expired

10. Who is the user that owns the data dictionary?

- a) SYS
- b) SYSTEM
- c) SYSBACKUP
- d) SYSDG
- e) SYSKM

11. Common users:

- a) Are same as local users
- b) Have SYSDBA privilege
- c) Exist in one pluggable database
- d) Span all pluggable databases within a container database

12. What is the wrong statement regarding the Authentication types?

- a) Operating system authentication (for administrators)
- b) Remote Desktop connection authentication
- c) Password authentication (for users)
- d) External authentication (for users)
- e) Global authentication (for users)

13. OS authentication takes precedence over password file authentication for privileged users?

- a) True
- b) False

14. Password authentication is encrypted and stored in?

- a) The Data Dictionary
- b) The Shared Global Area
- c) The Program Global Area
- d) The Shared Pool Area

15. What is the true statement for revocation of a system privilege?

- a) Won't be allowed
- b) Will be allowed only for SYS user
- c) Will be cascaded
- d) Won't be cascaded

16. What are the statements of types of user privileges?

- a) System privileges
- b) SQL DDL privileges
- c) Object privileges
- d) SQL DCL privileges

17. What is the wrong statement regarding the benefits of Roles?

- a) Easier privilege management
- b) Static privilege management
- c) Dynamic privilege management
- d) Selective availability of privileges

18. Schema Only Accounts that can be created without passwords?

- a) True
- b) False

19. Choose the correct statement that describe the following command:

SQL> create user app1 NO AUTHENTICATION;

- a) Adding privileges to the user app1
- b) Creation a User Account with password
- c) Creation a Schema Only Account
- d) Creation a Privileged User

20. You can't modify a default permanent tablespace, and temporary tablespace after creation a user?

- a) True
- b) False

Quiz answers – chapter 7	
1	d
2	c, d, e
3	b
4	b, c
5	b
6	a, b
7	b
8	e
9	c
10	a
11	d
12	b
13	a
14	a
15	d
16	a, c
17	b
18	a
19	c
20	b

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